

Modern Application Development II – Final Project Report

Hospital Management App – V2

1. Student Details

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About Me: I am a student in the IIT Madras BS Degree program. I have a strong interest in full-stack web development and application design. I enjoy creating applications that help solve real-world problems. I focus on using clean architecture, databases, and user-friendly interfaces.

2. Project Overview

Project Title: Hospital management system

The Hospital Management System (HMS) is a multi-user web application designed to streamline hospital operations. The system supports three main roles: Admin, Doctor, and Patient. Admins can manage departments, doctors, and user access, while also monitoring appointments and treatment records. Doctors can maintain their availability schedule, view appointments, and record diagnoses, prescriptions, and follow-up details. Patients can securely register, log in, and book appointments based on doctor specialization and availability.

3. Technologies and Frameworks Used

Technology / Library Purpose

Frontend: VueJS, Bootstrap

Backend: Flask (Python), Flask-Login

Database: SQLite

Templating: Jinja2 (entry point only via CDN)

4. Database Schema / ER Diagram:

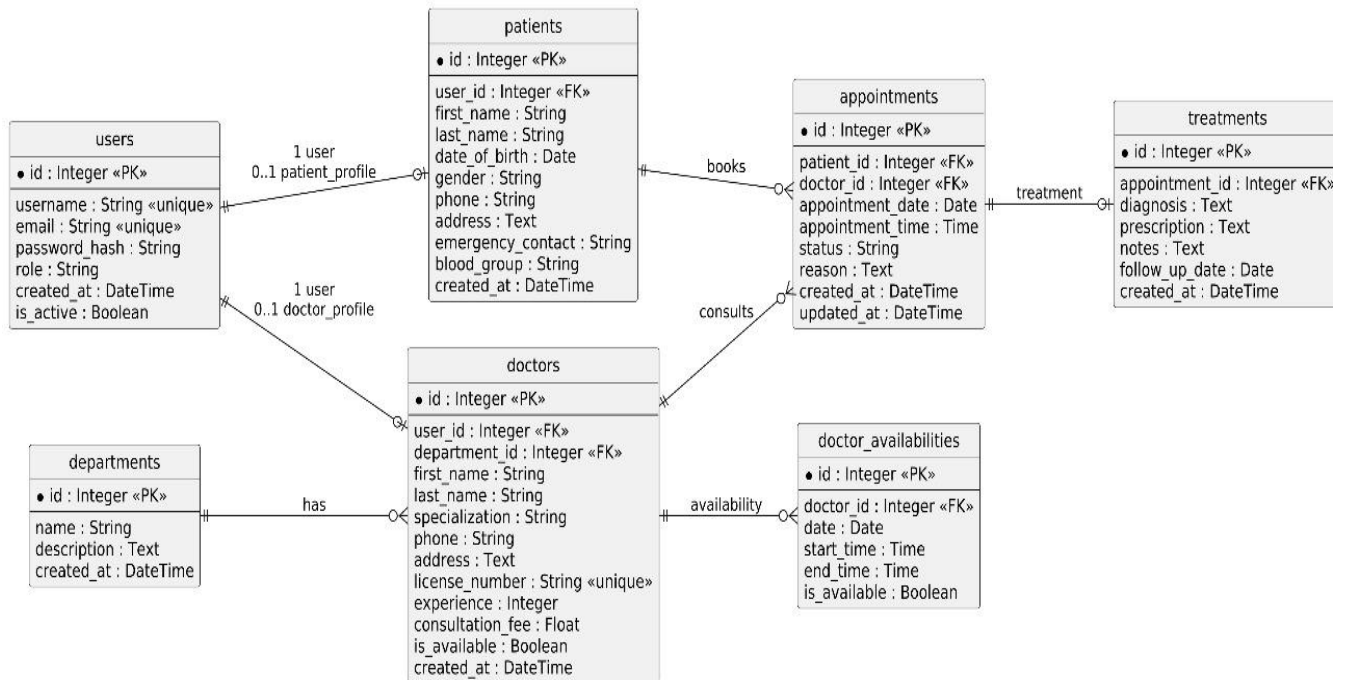
Key tables:

-Users (id, username, email, password_hash, role, is_active, created_at)

-Patients (id, user_id, first_name, last_name, phone, gender, blood_group, address, date_of_birth, emergency_contact, created_at)

-Doctors (id, user_id, department_id, first_name, last_name, specialization, license_number, consultation_fee, phone, experience, is_available, created_at)

- Departments** (id, name, description, created_at)
- DoctorAvailabilities** (id, doctor_id, date, start_time, end_time, is_available)
- Appointments** (id, patient_id, doctor_id, appointment_date, appointment_time, status, reason, created_at, updated_at)
- Treatments** (id, appointment_id, diagnosis, prescription, notes, follow_up_date, created_at)



5. Authentication & Access Control

- Role-based access control (RBAC) using Flask-Login
- Sessions manage user logins with clear role separation
- Passwords stored securely using hashing
- Unauthorized access to views is restricted based on role

6. UI/UX Highlights

- Built using VueJS with Bootstrap components
- Responsive design suitable for both desktop and mobile
- Validations implemented both on frontend and backend
- Clean and intuitive form layouts

7. Video Presentation

Drive Link:

https://drive.google.com/file/d/1uUDyBnwsQDjJcz1Z3f1sfXSmMsIKgCvz/view?usp=drive_link

Declaration

This report summarizes the project titled " Hospital Management App – V2", developed as part of the Modern Application